Garvit Sharma

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EXPERIENCE

AHA Solar Technologies Limited 09/2023 – Present
Software Development Engineer Ahmedabad, India

- Designed and scaled **full-stack** backend systems serving **3.2M+ users**, sustaining high performance and reliability at production scale.
- Architected **microservices** and backend services using **Node.js, Express.js, and MySQL** following a **layered architecture (controller–manager–data layer)** to improve modularity, maintainability, and testability.
- Built and shipped **RESTful APIs** with focus on **statelessness, idempotency, and fault tolerance**, supporting **15K+ daily requests** across high-throughput workflows.
- Centralized validation and middleware logic (**Joi-based**), reducing code duplication by **20%** and enforcing consistent request handling across **15+ service endpoints**.
- Optimized system performance using **Redis caching** and **query optimization**, reducing API response time by **60%** and lowering database load significantly.
- Automated **CI/CD** deployment and monitoring on **AWS EC2**, with secure file storage via **S3**, maintaining **99.9% uptime** across production operations.

PROJECTS

PMT-2.0 (AHASolar Project Management Tool) 02/2025 – Present

- Spearheaded scalable backend architecture enabling **20+ EPC companies** to automate solar project workflows, achieving **99.9% uptime** and reducing project completion time by **15%**.
- Designed and deployed **200+ REST APIs** handling **2K+ daily requests**, following **agile** development practices with stateless, modular, and maintainable service design.
- Integrated **JWT authentication** securing **200+ API endpoints** and enforcing role-based access control (**RBAC**) across all services.
- Restructured **MySQL schema** and applied **query optimization** across **20K+ records**, improving query performance by **35%**.
- Migrated data access patterns to **Redis caching**, cutting latency by **60%** and improving throughput under load.
- Consolidated **cron-based asynchronous pipelines** for workflow automation using **Docker** containers, reducing manual intervention by **30%**.

PMSG (Pradhan Mantri Surya Ghar) *(Completed Project)* 12/2023 – 02/2025

- Architected backend systems for India's **10M-user national-scale platform** offering **300 units/month** free electricity, guaranteeing fault-tolerant, scalable operations.
- Delivered **40+ APIs** across Vendor, DISCOM, and Consumer workflows handling **10K+ daily requests**; built **React.js** dashboards consuming these APIs for 3 user tiers.
- Accelerated onboarding by implementing **JWT**-based authentication and enforcing tiered permission controls across 3 user roles, serving **10M+ users**.
- Engineered **cron-based data pipelines** processing **450K+ records/month**, eliminating **30+ hours** of manual processing time per month.
- Established **input validation** pipelines maintaining **95% data accuracy** across **500K+ interactions**.

ACADEMIC PROJECTS

Indian Sign Language Recognition – *OpenCV, TensorFlow, CNN* 11/2022 – 06/2023

- Trained a **CNN-based deep learning model** on **25K+ annotated ISL images**, improving classification accuracy by **25%**; dataset published on Kaggle with **60+ downloads**.
- Accelerated detection pipeline achieving **<200ms latency**, enabling **500+ daily real-time gesture** recognitions.
- Achieved **84.24%** real-time accuracy validated with 50 participants; presented at **BITS Pilani Goa**; published with **Springer**.

EDUCATION

Vellore Institute of Technology 09/2022 – 05/2024
Master of Computer Applications (MCA) | ***CGPA: 8.3*** Chennai, Tamil Nadu

SKILLS

Languages: JavaScript, SQL
Backend: Node.js, Express.js, FastAPI, RESTful API Design, JWT Authentication, RBAC, Cron, Input Validation
Frontend: React.js, Redux
Cloud & DevOps: AWS (S3, EC2), Docker, CI/CD, Git, GitHub, GitLab
Databases: MySQL, PostgreSQL, Redis, Database Schema Design, Query Optimization
Practices: Agile, Layered Architecture, System Design, Performance Optimization